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Faisal D'Souza
Office of Science Technology Policy
2415 Eisenhower Avenue
Alexandria, VA 22314

Thank you for the opportunity to engage on this critical issue. Artificial intelligence (AI) is not only an emerging technology that will clearly impact our day-to-day lives but play a critical role in protecting our national security and promote economic competitiveness. Whichever country dominates AI will have significant military and economic advantages. While the United States is currently the leader in AI, as President Trump outlined in his January 23rd Executive Order (EO), we must act decisively to retain global leadership in this critical field.¹

Rapid expansion of AI and data centers will require dramatically more electricity to meet growing demand. [SemiAnalysis](#) used property records, power consumption data, and satellite imagery analysis to track U.S. data center developments and concluded that the amount of electricity needed to power U.S. data centers for AI will grow from 3 GW in 2023 to over 56 GW by 2028². To ensure we can dominate the AI of the future, we must have the necessary power to utilize it at the necessary scale while not increasing energy prices. If energy demands are not met, AI development could shift overseas. Energy abundance is essential to AI dominance, and we must take the necessary actions on energy policy to support it.

This dramatic increase in power demand cannot be met with any single source. In fact, we must ensure we have diversified energy production that reduces our reliance on any single feedstock. While the Trump administration has begun laying out its vision for achieving American Energy Dominance, one critical source must be in all energy planning: fusion energy.

At Helion Energy we are poised to make history during this administration with our seventh fusion prototype, Polaris, which will produce electricity from fusion for the first time in history. Following Polaris, we will construct the world's first commercial fusion power plant, backed by a power purchase agreement with Microsoft.³ We also have a customer agreement with Nucor to develop a 500-Megawatt plant to power one of their steel mills.⁴ These developments signal that fusion energy is no longer a distant vision—it is becoming a reality. This work is proceeding on schedule and in March 2025, we announced our intention to acquire permits for our desired location of our first power plant in Washington state.

¹ [Removing Barriers to American Leadership in Artificial Intelligence](#) (January 23, 2025)

² Special Competitive Studies Project, [How Bad Will the AI Power Crunch Be?](#) (September 5, 2024)

³ Helion Energy, Inc., [Announcing Helion's Fusion Power Purchase Agreement with Microsoft](#) (May 10, 2023).

⁴ Nucor Corporation, [Nucor and Helion to Develop Historic 500 MW Fusion Power Plant](#) (September 27, 2023).

Fusion energy isn't just a scientific milestone; it's a transformative force that can redefine the global energy landscape, fuel economic growth, and secure America's energy independence. We aren't the only country to recognize the potential of fusion energy. China is racing to dominate fusion, pouring billions into R&D as part of their broader strategy for global influence in advanced technologies and critical energy supply chains. The stakes couldn't be higher.

As global electricity demand rapidly rises, fusion energy presents an unprecedented opportunity to meet this growth with clean, affordable, reliable power. In the U.S. alone, electricity demand is projected to increase by nearly 30% by 2050, driven by electrification in transportation, industrial sectors, and the digital economy.⁵ And yet this is likely an underestimate given recent advances in AI and efforts to onshore critical industries.⁶ Globally, electricity consumption is expected to more than double during the same period, especially as developing nations industrialize and urbanize at a rapid pace. This surge in demand underscores the urgency of scaling up energy production, and fusion is poised to play a transformative role.

While the U.S. stands at the forefront of fusion innovation, we must recognize that global competition, particularly from China, is accelerating rapidly. China has made fusion a cornerstone of its national innovation strategy, aggressively investing in research, development, and manufacturing capabilities. This effort is not just aimed at civilian energy solutions but also has broader implications tied to its military-civilian fusion policy, where technological advancements in the civilian sector directly support military applications.

The United States must win the contest to dominating AI and controlling this emerging, and critical, industry. AI is poised to revolutionize numerous industries, creating new markets and driving significant economic growth. However, achieving energy abundance is a necessary condition for achieving AI dominance.

China is betting big on fusion. They are pouring billions into research and development, not just to lead in energy, but to power their aspirations for global dominance in AI, advanced manufacturing, and military innovation. As their grip on rare earth minerals tightens and their influence over critical energy supply chains grows, fusion is their endgame—a way to leapfrog the U.S. and redefine the global balance of power.

A key plank in the Chinese strategy to not only dominate fusion deployment, but to actively undermine the American fusion industry, is to exert control over the fusion energy supply chain. As we saw happen repeatedly over previous decades, they are positioning themselves to dominate the commercialization of American ingenuity and reap the benefits.

We cannot afford to fall into this same trap. We must lead.

⁵ Statista, [Projected Electricity Use in the U.S. 2022-2050](#) (June 28, 2024)

⁶ Washington Post, [Amid Explosive Demand, America is Running out of Power](#) (March 7, 2024)

Only America has the entrepreneurial spirit, scientific talent, and industrial base to turn fusion from a scientific breakthrough into a commercial reality. President Trump has the chance to deliver on this promise.

For America to reach its full fusion potential, the Trump administration should:

1. **Establish a National Fusion Lead:** Appoint someone with direct access to the Secretaries of the Department of Energy, Department of Commerce, and Department of Defense to encourage concrete actions toward fusion energy commercialization and deployment.
2. **Declare Fusion a National Security Priority:** as part of the national security strategy, launch a presidential task force to win the race on fusion by focusing on advancing the development and deployment of commercial fusion energy. The task force would cut through red tape, streamline environmental permitting, and create a clear regulatory framework for fusion.
3. **Public-Private Partnerships:** Engage with companies like Helion to expedite the commercialization of fusion. By leveraging the ingenuity of American industry, we can ensure that not only will the first commercial fusion plant be built in America and not in Beijing or Shanghai - with the right policies in place, we will also ensure we can rapidly deploy at scale such that Americans reap the benefits of a robust fusion economy.
4. **Secure Fusion Supply Chains:** Direct the Department of Commerce, and other relevant agencies, to activate existing programs to scale manufacturing of critical components, such as high-voltage capacitors and superconducting materials. This will reduce reliance on foreign imports and geopolitical competitors all while creating an American supply chain immune to external pressures.

The U.S. stands at a crossroads. We're on the brink of unlocking fusion energy, but only if the public and private sectors work together. The next few years will decide if America leads the world in fusion—or gets left behind. With bold action on supply chains and smart regulatory frameworks, fusion can revolutionize our energy future and cement U.S. leadership in the global energy race.

Thank you for the opportunity to participate in this important process. We hope to be a resource to you as we meet the dual goals of energy abundance and AI dominance.

Sincerely,

Jackie Siebens
Director of Public Affairs
Helion